

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Date: Sep 14, 2023

Pre-Tutorial Assignment 6

Marks: 2

1. Write the expression for the voltage $v_c(t)$ across the capacitor C for $t > 0$ (Figure 1). The numerical values of the circuit elements are as follows: $R_1 = 2\Omega$, $R_2 = 4\Omega$, $C = 10\mu F$ and $V_0 = 9V$.

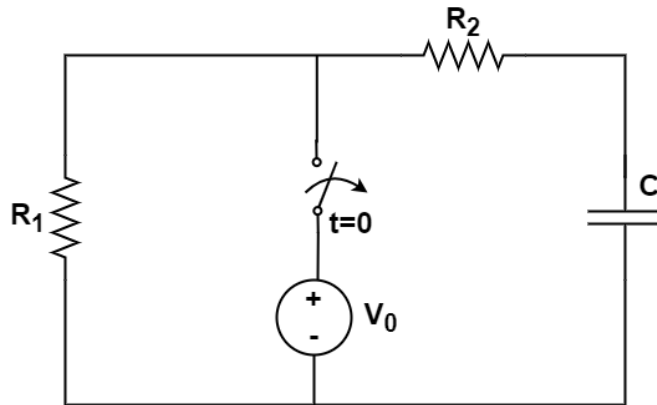


Figure 1: Electric circuit with an independent voltage source, a switch, resistors and a capacitor. The switch opens at $t = 0$.

2. Construct the following devices using a positive edge triggered JK Flip-Flop and logic gates.
 - (a) A positive edge triggered D Flip-Flop.
 - (b) A negative edge triggered T Flip-Flop.